Prathamesh Saraf

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EDUCATION

University of California, San Diego San Diego, USA
MS in Electrical & Computer Engineering (Intelligent Systems, Robotics, & Controls) *2022 - 2024*
Courses: Robot Motion Planning, Sensing and Estimation, Co-operative Control of Multi-Agent Systems, Statistical Learning, Convex Optimization, Linear Algebra, Non-Linear Controls, Stochastic Processes in Dynamic Systems

Birla Institute of Technology & Science, Pilani Hyderabad, India
B.E. Electronics & Instrumentation; Minor in Robotics & Automation *2017-2021*
Courses: Modern Control Systems, AI for Robotics

SKILLS

Programming & Scripting: Python, C/C++, RUST, Java, HTML, Verilog, PLC Programming Simulation,
Robotics & Control Tools: MATLAB, ROS, Simulink, Gazebo, Webots, PyBullet, MuJoCo, LabVIEW
CAD/Hardware: SolidWorks, SolidWorks Electrical, Fusion 360, AutoCAD, DraftSight, HDL, Allen-Bradley PLC, Benchtop

EXPERIENCE

Controls Engineer, **ASML** *June 2024 - Present*

- Designed and validated a closed-loop control system for the Energy Controller, applying system identification and optimal gain computation to enhance product performance and reliability for critical machine functions.
- Successfully led the end-to-end delivery of an open-loop control solution, coordinating with cross-functional teams to update requirements, design test cases, and perform rigorous unit/bench testing, ensuring timely feature deployment.
- Developed and deployed a 'Sensor Acquisition' Hardware-in-the-Loop (HIL) simulator on an FPGA, which improved validation capabilities for complex hardware interactions and facilitated robust regression testing on testbenches.
- Revived and upgraded a Python-based simulator to emulate hardware for High-NA machines, providing internal teams with a critical tool for rapid prototyping and validation, enhancing development efficiency.
- Provided critical escalation and integration support, rapidly debugging issues and delivering crucial FW/control patches in under 1 day, directly assisting proto engineers in feature bring-up and ensuring smooth system implementation.

Control Systems Engineering Intern, **Entegris Inc.** *June 2023 - Sept 2023*

- Developed electrical schematics and CAD models for Gas Purifier Systems, updated redlines, and Bills of Materials to ensure accurate documentation for manufacturing and customer support.
- Established Solidworks Electrical foundations and revamped Purifier Systems, directly reducing labor and rework costs through improved design and documentation processes.
- Rectified engineering errors and implemented changes that led to a 98% increase in throughput within two months, demonstrating strong problem-solving and optimization skills in a practical application.

Dynamics and Controls Intern, **N Robotics** *May 2023 - Sept 2023*

- Developed and implemented velocity and position-based gait transitions for autonomous bipedal navigation within a ROS-PyBullet framework, demonstrating practical application of robotic control for specific use cases.
- Trained a multi-clip policy for humanoid robot locomotion via motion imitation (MoCapAct) in MuJoCo, enhancing agility and efficiency in simulated robotic operations.

Research Assistant, **Stochastic Robotics Lab, Indian Institute of Science** 🔗 *June 2021 - May 2022*

- Led a team of 4 in developing a control solution for the 'Stoch' quadruped robot, including velocity-based gait transitions, a state estimator, and a whole-body impulse control + model predictive controller, significantly increasing robot stability and operational effectiveness in unknown environments and rough terrains.
- Verified and optimized controllers using OROCOS and ROS-Gazebo, culminating in a presentation to a space organization, showcasing technical solutions and fostering potential future collaborations.

Research Intern, **Multi-Agent Robotic Motion Lab, National University Singapore** 🔗 *Jan 2021 - Nov 2021*

- Designed a gradient descent-based inexpensive cost function to analyze the ground reaction forces and obtain the required body pose, thus empowering stable complex hexapod robot locomotion using Central Pattern Generators.
- Increased robot stability and robustness by ~40% in the simulation environment.

Electronics Subsystem Lead, **Hyperloop India** 🔗 *Aug 2019 - Oct 2020*

- Designed a distributed on-board electronics architecture and vehicle controller, with localized control loops for motor fine-tuning and executing an Emergency Braking Protocol for enhanced pod safety and operational reliability up to 98%.
- Developed a real-time Kalman filter for Hyperloop pod pose estimation, reducing acceleration RMSE by **91.7%** and velocity RMSE by **67.4%**, critical for precise vehicle control and motion planning.
- Engineered a comprehensive sensor fusion system, integrating IMUs, tachometers, optical encoders, and fiducial markers for accurate state estimation and feedback, validated through simulations over 120 discrete time-steps.

RELEVANT PROJECTS

Traffic Wave Dampening using Autonomous Vehicles | *Python*🔗 *Apr 2023 - Jun 2023*

- Modeled complex traffic flow dynamics using a state-space representation to validate Lagrangian control strategies for autonomous vehicles, proving a single AV's capacity to significantly dampen traffic waves and ensure system stabilizability.
- Implemented a Follower Stopper controller, reducing velocity standard deviation by **80.8%**, fuel consumption by **42.5%**, and excessive braking by **98.6%**, while concurrently increasing traffic throughput by **14.1%.**

Multi-Modal Sensor Fusion for Robotic Localization and Mapping | *Python*🔗 *Jan 2023 - Mar 2023*

- Implemented comprehensive Simultaneous Localization and Mapping solutions for autonomous robots, fusing IMUs, encoders, 2D LiDAR (up to 30m range), and stereo/RGBD camera data to construct 2D occupancy grids to track landmarks
- Optimized a 100-particle Particle Filter for differential-drive robot localization and an Extended Kalman Filter (EKF) for visual-inertial SLAM, ensuring robust and accurate pose tracking.
- Validated high-fidelity sensor processing pipelines, encompassing IMU calibration, 10Hz yaw rate filtering, and gradient descent optimization for quaternion-based orientation tracking for precise vehicle control and panoramic mapping.

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A Convolutional Neural Network Approach Towards Self-Driving Cars | *C++, Python* *Sept 2018 - Mar 2019*

- Implemented a Convolutional Neural Network (CNN) for Level 2 autonomous vehicle steering control, achieving a Mean Squared Error (MSE) loss of **0.0003798** after 30 epochs by directly mapping camera input to steering commands.
- Integrated a robust obstacle avoidance system with ultrasonic sensors and the RRT*-Connect algorithm, enabling autonomous lane changes based on obstacle velocity tracking within a **20cm** detection threshold.
- Engineered the vehicle's control architecture, integrating a Raspberry Pi for image processing and an Arduino Mega for motor control, resulting in an **86.43%** autonomy rate during testing on the CARLA open-source driving simulator while adhering to ISO26262 functional safety principles.

PUBLICATIONS / PROJECT REPORTS

1. **P. Saraf,** M. Shaikh, M. Phan, "*Convex Optimization in Legged Robots*," Project Report MAE 227
2. **P. Saraf,** "Traffic Wave Dampening using Autonomous Vehicles," Project Report MAE 247
3. **P. Saraf,** Y. Jangir, R. N. Ponnalagu, "*Implementation and Testing of Force Control on a Spherical Articulated Manipulator*," IEEE International Conference on Mechatronics and Automation (ICMA), 2022
4. **P. Saraf,** et al, "*Terrain Adaptive Gait Transitioning for a Quadruped Robot using Model Predictive Control*," IEEE International Conference on Automation and Control (ICAC), 2021
5. **P. Saraf,** et al, "*Onboard Electrical, Electronics, and Pose Estimation System for Hyperloop Pod Design*," IEEE 7th International Conference on Control, Automation and Robotics (ICCAR), 2021
6. **P. Saraf,** R. N. Ponnalagu, "*Modeling and Simulation of a Point-to-Point Spherical Articulated Manipulator Using Optimal Control*," IEEE International Conference on Automation, Robotics, and Applications (ICARA), 2021
7. **P. Saraf,** M. Gupta, and A.M. Parimi, "*A Comparative Study Between a Classical and Optimal Controller for a Quadrotor,*" IEEE 17th India Council International Conference (INDICON), 2020
8. A. Agnihotri, **P. Saraf,** and K.R. Bapnad, "*A Convolutional Neural Network Approach Towards Self-Driving Cars*," IEEE 16th India Council International Conference (INDICON), 2019

TEACHING / LEADERSHIP

1. **Electrical Subsystem Lead**, Hyperloop India, BITS Pilani *Aug 2019 - Oct 2020*
2. **Teaching Assistant -** Control Systems Laboratory, BITS Pilani *Jan 2020 - May 2020*

Provided direct technical support and troubleshooting assistance during lab sessions.

3. **Robotics Mentor -** Student Mentorship Program, BITS Pilani *Aug 2019 - Dec 2019*

Mentored students in robotics projects, offering technical guidance, problem-solving strategies, and encouragement to develop their skills and complete challenging tasks.